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Palaeomagnetism of the Macaronesian Insular Region: The Cape Verde Islands

N. D. Watkins, A. Richardson and R. G. Mason

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Summary

Palaeomagnetic pole positions have been determined for a collection of igneous rocks, comprising nearly five hundred samples, from the Cape Verde Islands of Santa Antao, Sao Vicente, Sao Nicolao and Sao Tiago. Limited data from the islands of Sal, Maio and Fogo are also presented. Stratigraphic control suggusts that the lavas are overwhelmingly Miocene in age on Sao Tiago and Sao Nicolao. Similarity in the palaeomagnetic pole positions indicates that Miocene lavas are also dominant on Santa Antao and Sao Vicente.

Substantial areas within two of the islands are of reversed polarity only, suggesting either a rapid extrusion rate, or the existence of a long reversed polarity epoch during the Miocene period. The palaeomagnetic pole positions for each island are close to the present geographic pole, excluding the possibility of Post-Miocene differential crustal spreading (or rotation about a vertical axis) in this part of the Atlantic. The palaeomagnetic pole position for the entire survey is consistent with the Miocene geographic pole being removed from, but close to, the present geographic pole; and is in harmony with the European polar wandering curve.

Introduction

This is the fourth in a series of papers describing the palaeomagnetic pole positions obtained from Atlantic islands (Watkins *et al.* 1966a, b; Richardson & Watkins 1967). These include the Canary Islands and Madeira, which together with the Selvagens, the Azores and the Cape Verde Islands, constitute the Macaronesian insular group. The palaeomagnetic surveys are forming the bases of diverse geological surveys, some of which have already been reported (Abdel-Monem *et al.* 1967).

Recent evidence consistent with crustal spreading (Vine 1966; Heirtzler & Hayes 1967) and continental drift in the Atlantic (Hurley *et al.* 1967) if valid, would suggest that data pertinent to differential crustal spreading, or rotation about a vertical axis (Pitman & Heirtzler 1966, Fig. 5 (b)), in this part of the east Atlantic is worthy of consideration. Thus, although incapable of detecting latitudinal movement in Tertiary rocks, palaeomagnetic methods in low latitudes can provide evidence relevant to major possible crustal tectonics in the Atlantic, in addition to delineating the local stratigraphic potential of any existing polarity variations within an island. At present, palaeomagnetic polarity measurements appear to offer the most promising stratigraphic tool in volcanic areas, because of the general lack of stratigraphically useful sediments and fossils. The local stratigraphic value would be low, however, if only a single polarity were recorded by the igneous material.

Previous Geological Studies

The Cape Verde archipelago (Fig. 1), despite its nodal position on shipping routes, and its pertinence to the genesis of the Atlantic basin, has been only rarely visited by geologists.

Great between-island geomorphological contrasts characterize the islands. The very deeply dissected western slopes of Santa Antao are similar to parts of western Sao Nicolao and northern Sao Tiago. Broader but still deep valleys are found in Sao Vicente, northern and central Sao Tiago and Brava. Erosion has drastically reduced relief on Sal and Maio, where marked unconformities exist. Younger, rounded topography is seen on any large scale only in eastern Santa Antao, and on Fogo, an island which is a single massive volcano rising to 2830 m above sea level.

In common with almost all Atlantic islands, the islands are composed dominantly of basic volcanic materials with only rare ultrabasic or acid igneous rocks. Bebiano (1932) has published by far the most comprehensive geological (but not stratigraphic) maps, which are included in Fig. 2(a)–(f). Berthios (1950) has summarized much of the petrological and chemical results so far obtained. As well as presenting a short résumé of the geological literature of the islands, Part (1950a) has made additional petrological and chemical analyses, and has described the quartz monzonites of Sao Vicente and Sao Tiago (1950b). The most detailed petrological and chemical study so far made is restricted to the island of Fogo (Torre de Assuncao 1954).

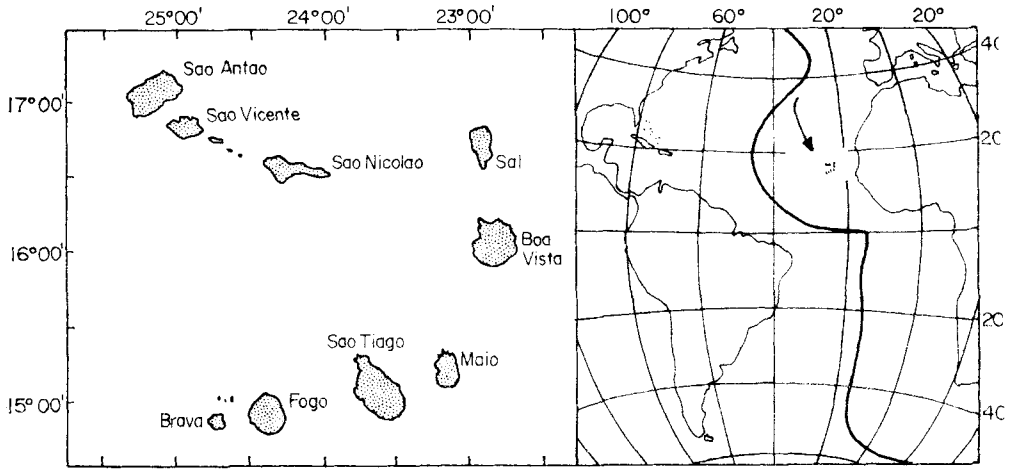


FIG. 1. Sketch map showing location of the major islands of the Cape Verde archipelago.

Considering their small volume compared to the volcanics, the sediments of the Cape Verde Islands have been relatively well studied. Their fossil content provides upper limits to the ages of the outcrops on some of the islands. Possible Portlandian (Upper Jurassic) materials (Sousa Torres & Pires Soares 1946) underly Aptian (Lower Cretaceous) carbonates and cherts (Stahlecker 1935), which are also believed to exist in Boa Vista and Sal (Pires Soares 1948), beneath major unconformities. Senonian (Upper Cretaceous) and Lutetian (Eocene) fossils occur at 250 m above sea level on Mount Focinho in Sao Nicolao, within carbonates interdigitated between basalts (Bebiano & Pires Soares 1951). With the exception of Santa Antao and Fogo, Miocene sediments appear to be indigenous to all islands (Sousa Torres & Pires Soares 1946, 1950; Pires Soares 1956), with younger sediments also being widespread. Pires Soares (1954) has classified these sediments according to their suspected origin. Part (1951) has examined the petrology of some metamorphosed examples of the Miocene carbonates on Sao

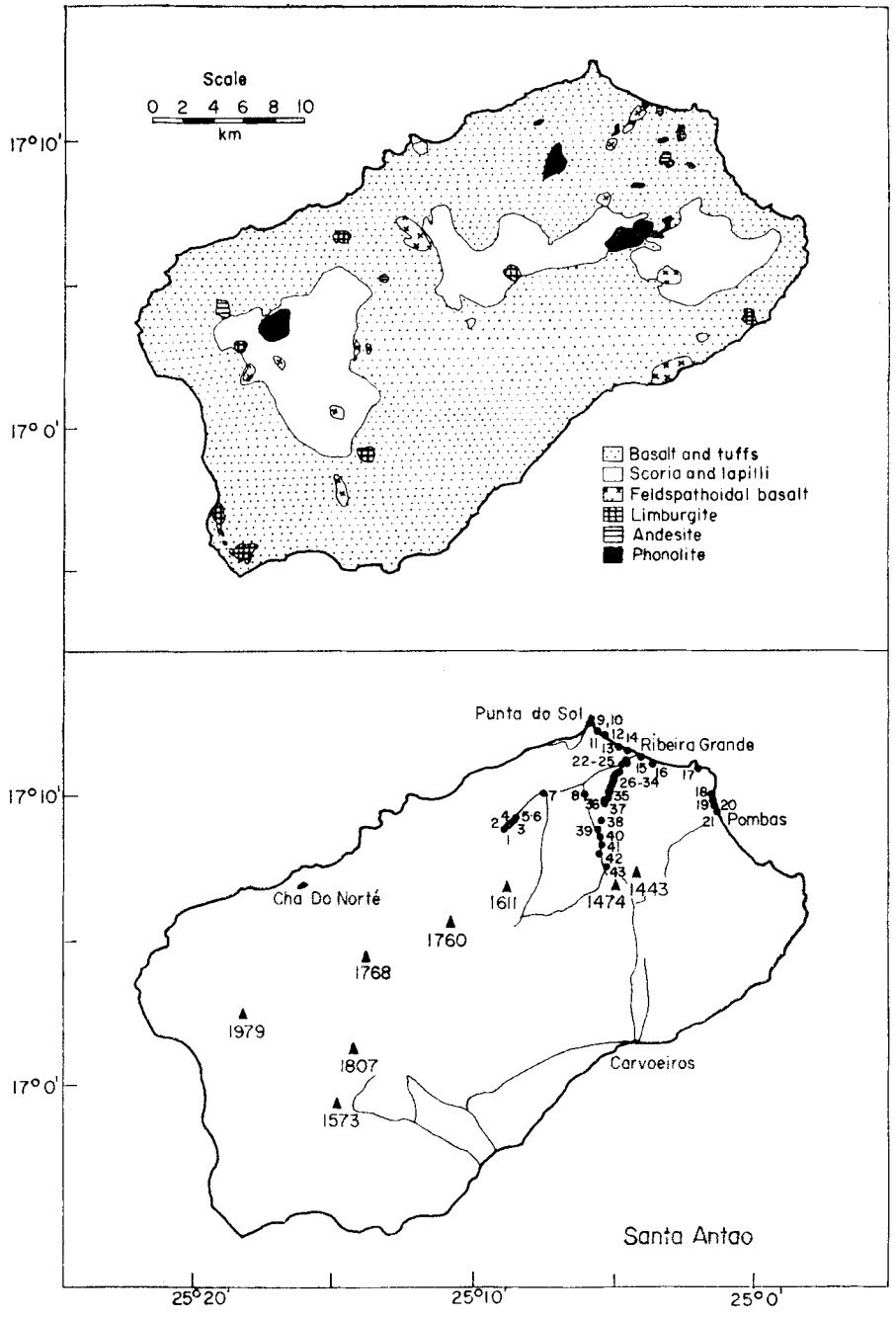


FIG. 2 (a)

Tiago, from close to Praia. Hausen (1956, p. 242) has summarized much geochemical data from Atlantic islands, and shows that the Cape Verdes are easily the most alkali rich of the Atlantic. McBirney & Gass (1967) beleive that this property is inherent to islands well separated from the mid-Atlantic ridge, reflecting magma genesis zones quite different from the areas close to the ridge.

Bebiano (1932) has suggested that the linear distributions of the islands may be

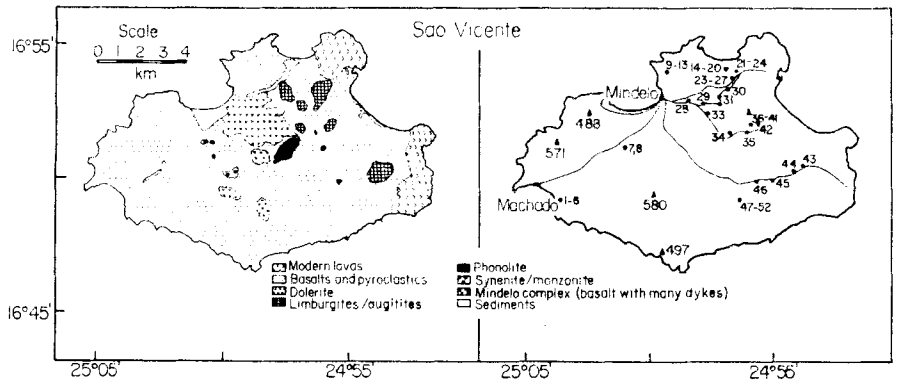

Scale
0 1 2 3 4
km
Sao Vicente
16°55'
16°45'
25°05'
24°55'
25°05'
24°56'
Mindelo
Machado
488
571
580
497
Modern lavas
Basalts and pyroclastics
Dolerite
Limburgites/augitites
Phonolite
Syenite/monzonite
Mindelo complex (basalt with many dykes)
Sediments

FIG. 2(b)

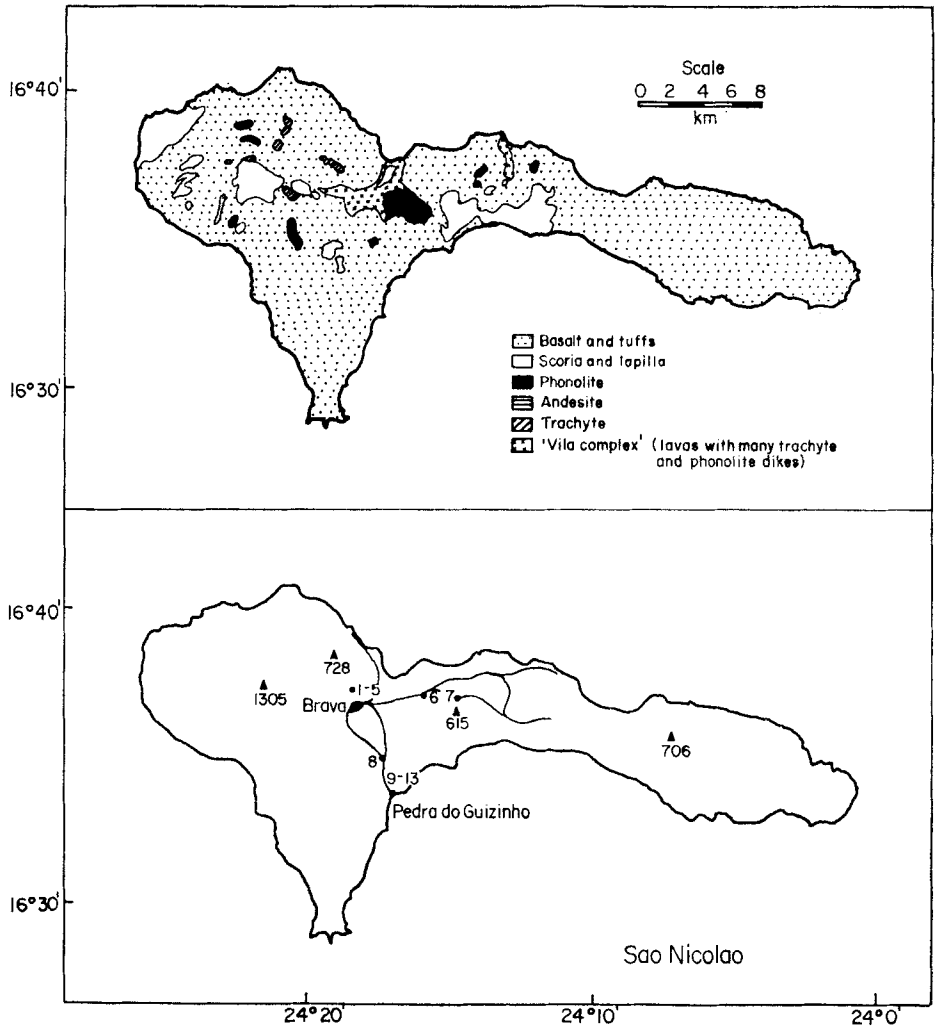

Scale
0 2 4 6 8
km
16°40'
16°30'
Basalt and tuffs
Scoria and lapilla
Phonolite
Andesite
Trachyte
'Vila complex' (lavas with many trachyte and phonolite dikes)
728
1305
Brava
615
706
Pedra do Guizinho
Sao Nicolao
24°20'
24°10'
24°0'

FIG. 2(c)

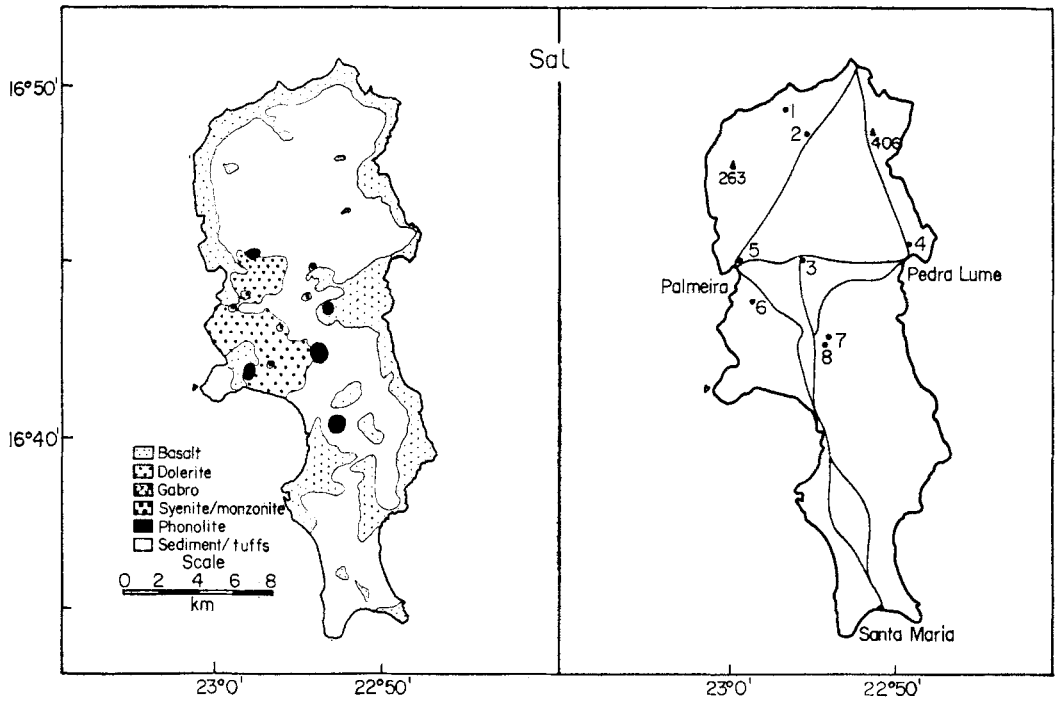

Sal
16°50'
16°40'
Basalt
Dolerite
Gabro
Syenite/monzonite
Phonolite
Sediment/ tuffs
Scale
0 2 4 6 8
km
23°0'
22°50'
1
2
406
263
5
3
4
Palmeira
Pedra Lume
6
7
8
Santa Maria
23°0'
22°50'

FIG. 2(d)

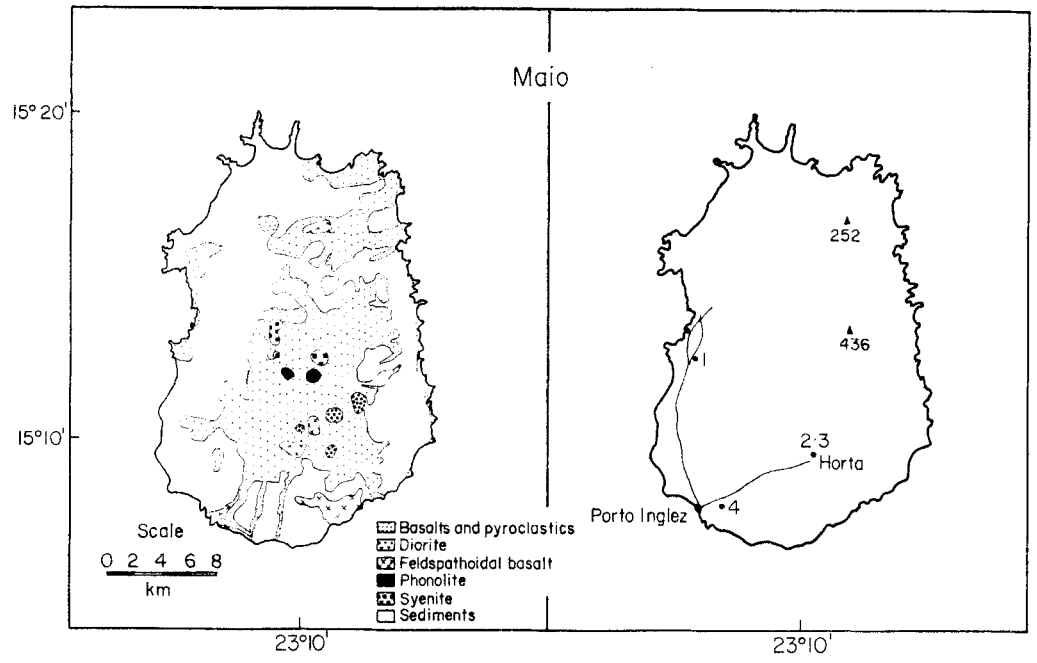

Maio
15° 20'
15°10'
Scale
0 2 4 6 8
km
Basalts and pyroclastics
Diorite
Feldspathoidal basalt
Phonolite
Syenite
Sediments
23°10'
252
436
1
2·3
Horta
Porto Inglez
4
23°10'

FIG. 2(e)

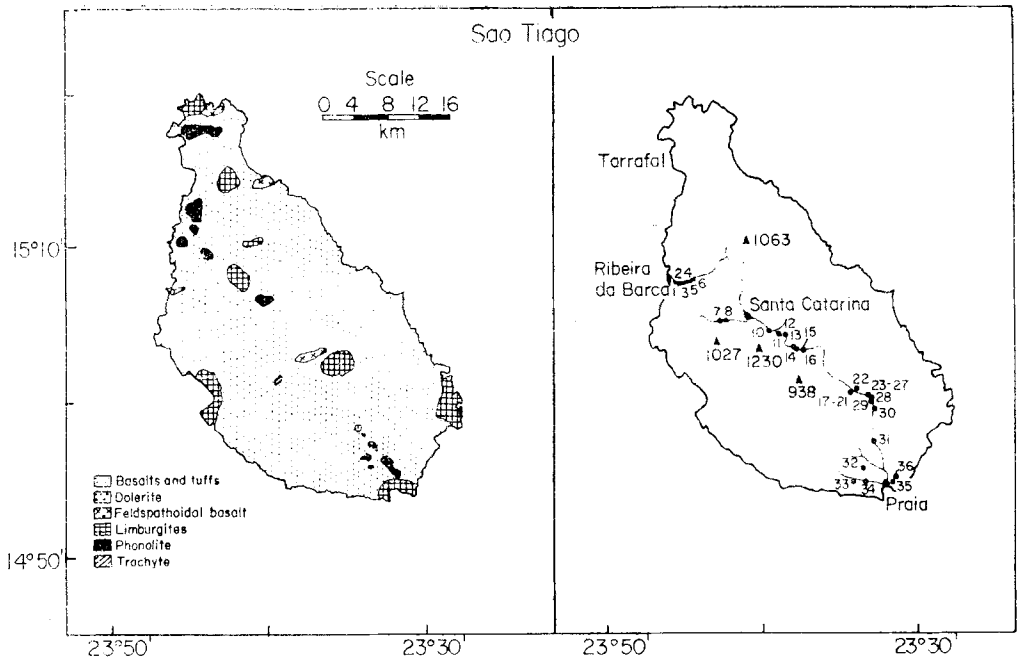


FIG. 2(f)

Fig. 2. Generalized surface geology and sample site locations for (a) Santa Antao, (b) Sao Vicente, (c) Sao Nicolau, (d) Sal, (e) Maio and (f) Sao Tiago. Each surface geological map is after Bebiano (1932). Each sample location map shows major roads or tracks, and settlements. Major topographic features are indicated by solid triangles, with the height given in metres.

controlled by major faults. Part (1950a) has summarized some other speculations about the origin of the islands. Friedlander (1913) believes that the orogeny which created the Atlas mountains also gave rise to the Cape Verdes. He has described raised beaches at 280, 350, and 500 m above sea level. Wilson (1963a) believes that large uplift on oceanic islands may be a manifestation of an association with an oceanic ridge system.

The age of the Cape Verde islands, when compared to those of other islands, has been used by several authors (Wilson 1963b; Vogt & Ostenso 1967) to be indicative of crustal spreading, implying therefore that the Mesozoic carbonates originated when the Atlantic Basin was very narrow. Menard (1965) has commented on Wilson's (1963b) data, and McBirney & Gass (1967) use petrological arguments in opposing the implication that non-axial volcanic islands orginate at the axial zones.

Experimental methods

Field methods. The palaeontological work referred to above provides maximum ages for the sampled islands. Major unconformities separate the sample basalts from the fossiliferous sediments on Maio and Sal, and 'complexes' (zones dominated by dikes and sills) underly the sampled areas in Sao Vicente, and Sao Nicolao. The sampled bodies would appear to be overwhelmingly Miocene in age, although it is probable that a limited number of younger and older lavas are also included in the collection. Their palaeomagnetic contribution would tend to average out to a Miocene observation, but increase the confidence parameter size. Wherever possible, sequences

of successive lavas have been collected. The samples are, however, mostly confined to roadcuts, because of the general inaccessability, as well as very weathered nature of the natural outcrops.

A petrol-powered drill (Doell & Cox 1965) has been used to take at least three cores, of 2·5 cm diameter and average length 10 cm, from each body sampled. The cores were oriented to an accuracy of 3° of arc, in geographic coordinates, while still attached to outcrop. Some of the lavas exhibit dips up to five degrees, but no evidence shows that this is tectonic in origin. Sample locations are included in Fig. 2(a)–(f) alongside Bebiano's (1932) surface geological maps.

Laboratory methods. Each core was sliced into a specimen for measurement of magnetic properties, and into specimens for petrological, geochemical, and isotope studies, which will eventually be reported elsewhere.

An astatic magnetometer with a maximum sensitivity of 10^{-6} e.m.u. per mm scale deflection was used to measure the natural remanent magnetism (NRM) of each specimen, which was remeasured twice after the application of alternating magnetic fields of 100 and 200 Oe, respectively. The mean NRM direction for each body is computed by using one direction from each specimen, resulting from either the 100 or 200 Oe treatment, so that a minimum scatter NRM is calculated. Watkins & Richardson (1967) have discussed the reasons for this procedure.

Results

The mean NRM, together with additional Fisher (1953) statistics, are shown for each body in Table 1. Sample numbers correspond to the locations shown in Fig. 2(a)–(f). As indicated, some of the sampled bodies have a NRM which is too poorly defined (Vincenz & Bruckshaw 1960) to be considered a reliable indicator of the ancient geomagnetic field direction. The data for each island are presented as equal-area projections and as virtual geomagnetic poles in Fig. 3(a)–(f). Combining the NRM for each body provides the palaeomagnetic pole for Santa Antao, Sao Vicente, Sao Nicolao and Sao Tiago, and a mean virtual geomagnetic pole for the samples from Sal and Maio, in Fig. 4. (It is considered that sufficient samples were collected from the first four islands to minimize secular variation and, therefore, to provide a reasonable estimate of the spin axis position, but too few samples could be collected from Sal and Maio.) Ninety-five per cent ovals of confidence are omitted for clarity. Since only one lava was sampled on Fogo, presentation of the resulting data is confined to the tables.

Discussion

Discussion is most appropriately made about the significance of the polarity variation in each island, prior to consideration of the combined data for each island, and the result of combining all data for the entire survey.

(i) *Polarity variation in each island*

Santa Antao. (Figs 2(a) and 3(a)). The bodies sampled range stratigraphically from lavas and dikes deep within the great canyon of Ribeira Grande (units 8–13) to the capping phonolites in the north-central area at over 100 m above sea level (unit 43). The youngest rocks, which include welded tuffs, occur along the northeastern coast (units 13–16). Generally, the lavas sampled are massive, ranging between 10 and 50 m thick, but the dikes average only about 5 m in width.

Despite the large range in elevation of those sampled bodies which are away from the coastline, only a single polarity (reversed) is encountered, suggesting either a long reversed Miocene geomagnetic polarity epoch, or very rapid accumulation of about

Table 1

Cape Verde Islands palaeomagnetic data
(*Each unit, by island*)

(a) Santa Antao

No.	Type	N	D	I	K_0	K_{100}	K_{200}	K_{max}	θ_{63}	ϕ'	θ'	Polarity
1	E	3	183·7	−32·1	405·0	340·8	339·7	344·0	4·37	241·3	−86·5	R
2	I	3	177·8	+ 3·2	29·9	40·4	102·5	171·6	6·18	341·8	−71·1	R
3	I	3	186·6	−18·0	856·7	1231·1	502·0	1253·7	2·29	295·2	−79·8	R
4	I	3	163·2	−13·0	285·8	487·7	226·8	562·1	3·42	33·9	−70·5	R
5	I	3	186·3	−16·1	62·0	91·0	151·6	152·5	6·56	299·8	−79·1	R
6	E	3	165·4	− 2·2	3·5	6·1	26·5	27·2	15·54	18·2	−68·4	R
7	E	3	176·3	−15·8	137·3	147·0	137·9	151·8	6·57	356·9	−80·1	R
8	E	3	184·7	−26·0	121·7	279·1	507·6	630·3	3·23	281·7	−84·3	R
9	E	3	17·5	+17·4	7·6	6·6	6·1	6·9	30·93	88·7	+71·1	N
10	E	3	178·1	−26·6	27·6	31·7	40·0	41·2	12·61	4·9	−86·4	R
11	E	3	195·3	− 7·7	198·0	572·9	275·7	586·2	3·35	284·9	−70·2	R
12	E	3	229·6	−47·0	4·0	1·8	1·8	4·1	—	—	—	*
13	E	3	359·5	+13·6	13·0	16·4	16·3	17·4	19·41	157·4	+79·7	N
14	E	3	183·9	−22·5	208·0	220·7	399·6	866·6	2·75	299·9	−83·3	R
15	E	3	6·7	+21·0	418·2	360·2	21·4	380·0	4·15	108·4	+80·9	N
16	E	3	3·9	+66·7	1·5	2·0	1·1	2·4	—	—	—	*
17	E	3	350·7	+13·2	139·2	171·3	182·9	197·7	5·76	196·9	+76·1	N
18	E	3	188·3	−23·8	107·0	1137·6	950·0	1326·2	2·22	274·3	−80·7	R
19	E	3	182·6	−18·6	5·3	480·0	312·3	2025·6	1·80	316·1	−81·9	R
20	E	3	184·9	−42·7	14·6	94·5	127·9	161·4	6·38	185·1	−81·2	R
21	E	3	184·5	−44·8	87·3	190·9	306·6	319·8	4·53	178·5	−79·9	R
22	E	3	352·0	+29·4	89·5	324·7	208·5	352·5	4·31	235·2	+82·2	N
23	E	3	347·7	+20·9	60·4	61·1	68·8	123·4	7·29	218·4	+76·4	N
24	E	3	358·2	+22·8	2·5	2·3	2·6	2·6	—	—	—	*
25	E	3	160·9	−36·3	249·6	306·9	325·8	371·8	4·20	77·2	−71·7	R
26	E	3	192·2	−24·7	84·7	320·0	235·9	370·3	4·21	263·1	−77·5	R
27	E	3	196·8	−24·5	87·4	674·1	513·9	2578·5	1·60	257·7	−73·2	R
28	E	3	164·8	+15·7	12·3	12·1	18·0	18·8	18·70	6·9	−60·6	R
29	E	3	177·9	− 5·9	183·3	351·9	328·0	960·3	2·61	343·2	−75·6	R
30	E	3	173·3	−20·1	80·3	373·3	59·4	373·6	4·19	19·4	−80·6	R
31	E	3	182·2	− 9·3	20·4	100·1	239·4	316·9	4·55	324·9	−77·3	R

Table 1 (*continued*)

No.	Type	N	D	I	K_0	K_{100}	K_{200}	K_{max}	θ_{63}	ϕ'	θ'	Polarity
32	E	3	171·6	−13·5	3·3	131·5	172·0	247·9	5·14	14·2	−76·8	R
33	E	3	189·7	−20·3	1067·9	127·7	109·3	304·5	4·64	279·0	−78·4	R
34	E	3	186·5	−26·1	4·34	67·3	130·6	257·8	5·05	272·9	−82·8	R
35	E	3	172·9	−22·0	8·9	243·8	133·1	312·7	4·58	25·8	−81·0	R
36	E	3	280·4	− 2·9	7·0	1·3	1·3	9·4	26·43	60·4	+ 9·5	T
37	E	3	179·2	− 8·9	149·6	377·4	265·1	384·0	4·13	338·5	−77·3	R
38	E	3	124·0	+85·9	6·1	1·0	1·0	116·3	7·51	341·9	+12·5	T
39	E	3	340·8	+15·5	60·1	29·0	72·0	250·6	5·12	220·9	+69·1	N
40	E	3	186·2	−52·6	1·0	148·8	100·9	207·1	5·63	173·0	−73·1	R
41	E	3	185·1	−38·2	61·6	77·4	117·5	118·4	7·44	202·6	−83·5	R
42	E	3	187·2	−36·5	14·2	38·4	126·7	142·2	6·79	219·3	−82·6	R
43	I	3	196·1	−42·6	245·8	921·2	496·4	934·2	2·65	215·7	−73·2	R

(b) Sao Vicente

No.	Type	N	D	I	K_0	K_{100}	K_{200}	K_{max}	θ_{63}	ϕ'	θ'	Polarity
1	E	3	40·8	+36·9	5·2	24·3	1·3	45·1	12·06	53·0	+51·3	N
2	E	3	198·1	−49·2	2·7	1·6	7·4	8·3	28·12	203·2	−68·8	R
3	E	3	236·1	−82·6	1·6	1·1	1·2	12·9	22·56	168·2	−24·5	T
4	E	4	192·4	−12·8	4·9	5·6	21·7	27·1	15·57	284·2	−74·0	R
5	E	3	163·7	−15·5	100·8	451·2	259·4	623·5	3·24	37·5	−71·7	R
6	E	3	164·6	−19·4	171·0	407·6	433·2	878·8	2·73	42·2	−73·5	R
7	I	5	207·0	+ 3·1	11·9	13·9	32·5	33·8	13·94	277·1	−57·6	R
8	I	5	187·6	−19·6	4·7	12·9	59·5	84·0	8·84	286·8	−80·0	R
9	E	3	169·8	−54·7	4·6	5·9	1·9	6·3	32·30	130·5	−69·5	R
10	E	3	195·4	−41·8	524·6	702·7	266·2	707·6	3·05	216·1	−73·9	R
11	E	3	190·5	−30·7	230·9	311·1	379·4	380·5	4·15	245·6	−80·0	R
12	I	3	224·1	−21·3	13·6	8·0	10·0	19·7	18·24	246·9	−46·9	R
13	E	3	200·4	−24·1	2·8	304·8	846·9	1201·3	2·34	254·5	−69·8	R
14	E	3	204·0	−34·7	128·5	161·6	159·7	184·1	5·97	235·9	−67·1	R
15	E	3	196·4	−24·1	141·9	213·8	207·5	253·5	5·09	257·9	−73·6	R
16	E	3	206·2	−29·5	130·0	114·6	128·1	133·6	7·01	243·7	−64·9	R

Table 1 (*continued*)

No.	Type	N	D	I	K_0	K_{100}	K_{200}	K_{max}	θ_{63}	ϕ'	θ'	Polarity
17	E	3	201·4	−24·5	43·3	48·0	56·3	57·6	10·67	253·2	−68·9	R
18	E	3	205·8	−25·5	56·0	104·5	189·4	200·9	5·72	249·4	−64·9	R
19	E	3	208·1	−31·8	104·8	107·0	101·5	107·4	7·82	240·1	−63·1	R
20	E	3	215·2	−35·9	32·1	559·5	1022·9	2020·9	1·80	234·3	−56·6	R
21	E	3	343·7	− 3·0	283·5	298·0	262·9	381·1	4·15	17·8	+65·6	N
22	E	3	342·0	− 1·2	185·5	394·4	115·5	437·1	3·87	22·2	+65·1	N
23	E	3	1·9	+17·2	63·9	151·0	63·7	163·7	6·33	141·8	+81·7	N
24	E	3	20·3	+44·9	3·3	32·3	70·4	75·9	9·30	34·5	+68·9	N
25	E	3	11·7	+28·0	175·5	302·6	296·4	312·6	4·28	73·6	+78·6	N
26	E	3	10·2	+16·0	71·9	116·4	39·3	117·0	7·49	105·2	+76·8	N
27	E	3	355·9	+37·8	151·7	27·7	58·8	61·6	10·32	293·4	+84·2	N
28	E	3	175·9	−61·2	1·5	1·3	1·3	1·7	—	—	—	*
29	E	3	178·9	−17·3	17·4	173·0	231·0	243·0	5·20	342·5	−81·9	R
30	E	3	13·3	+18·4	441·9	346·2	22·9	1491·5	2·10	93·4	+75·1	N
31	E	3	186·9	−23·6	349·9	301·2	257·4	374·3	4·19	278·6	−82·0	R
32	E	3	350·7	+ 5·2	211·7	258·0	287·1	306·9	4·62	188·7	+73·0	N
33	E	3	166·4	−16·5	2·4	44·5	80·7	80·8	9·01	34·0	−74·3	R
34	E	3	342·5	+ 2·8	12·3	54·6	17·3	54·6	10·96	204·8	+66·8	N
35	I	3	1·8	− 2·6	6·5	50·0	54·5	56·6	10·77	149·3	+71·7	N
36	E	3	185·1	−44·3	9·9	700·0	226·4	1478·1	2·11	181·7	−79·7	R
37	E	3	183·6	−41·9	254·0	3004·9	1422·3	3620·3	1·35	179·2	−82·0	R
38	E	3	183·7	−35·5	424·6	3·5	204·7	612·0	3·27	207·2	−85·5	R
39	E	3	200·5	−16·9	119·3	2·2	3·0	3·1	—	—	—	*
40	E	3	196·8	−29·6	6·4	40·5	131·5	143·5	6·76	246·4	−73·8	R
41	E	3	203·5	−27·2	100·7	354·0	174·2	407·6	4·01	284·0	−67·3	R
42	E	3	251·8	−16·8	2·1	1·4	1·5	3·4	—	—	—	*
43	E	3	2·2	+32·4	32·2	9·2	81·1	91·7	8·46	46·9	+87·7	N
44	E	3	13·9	+14·3	227·9	249·5	72·0	581·4	3·36	98·6	+73·4	N
45	I	3	310·6	+42·7	98·8	16·7	18·0	35·8	13·53	263·4	+43·4	N
46	E	3	92·6	−12·2	3·7	2·4	2·8	2·8	—	—	—	*
47	E	3	171·1	+ 1·5	160·3	110·8	44·6	113·4	7·61	2·4	−70·3	R
48	E	3	169·3	−14·2	180·4	137·3	37·0	538·1	3·49	23·4	−75·8	R
49	E	3	173·9	−20·8	3·6	299·0	337·1	338·2	4·40	19·8	−81·5	R
50	E	3	178·0	+ 1·4	1·0	80·2	96·8	1·0	—	—	—	*
51	E	3	153·0	+58·5	2·7	2·3	3·8	3·8	—	—	—	*
52	I	3	343·3	+52·5	68·7	48·7	31·8	63·8	10·14	295·1	+67·9	N

Table 1 (*continued*)

(c) Sao Nicolau

No.	Type	N	D	I	K_0	K_{100}	K_{200}	K_{max}	θ_{63}	ϕ'	θ'	Polarity
1	E	3	190·2	−23·1	857·0	686·8	812·3	1006·8	2·55	269·2	−79·1	R
2	E	3	182·9	−14·9	119·6	166·9	172·2	203·3	5·68	317·8	−80·5	R
3	E	3	189·3	−14·5	664·7	1216·1	1069·1	1496·9	2·09	290·2	−77·1	R
4	E	3	184·5	−14·9	4·4	103·8	85·9	106·2	7·86	309·3	−80·0	R
5	E	3	215·0	−22·9	2·5	2·6	2·7	2·8	—	—	—	*
6	E	3	352·3	+56·7	173·7	234·6	173·8	233·4	5·30	318·9	+68·2	N
7	E	3	347·4	+28·0	144·6	192·4	296·4	341·4	4·38	239·5	+77·7	N
8	E	3	4·8	+14·1	415·1	123·2	230·0	874·7	2·74	128·6	+79·4	N
9	E	3	184·6	−28·3	254·7	181·2	248·3	274·1	4·89	264·0	−85·4	R
10	E	3	178·8	−28·5	5·9	391·2	100·3	569·0	3·40	16·3	−88·2	R
11	E	3	173·7	−30·5	3·0	74·1	111·0	146·5	6·69	65·1	−84·0	R
12	E	3	186·5	−30·3	301·1	193·3	326·1	340·4	4·39	247·8	−83·7	R
13	E	3	179·9	−30·2	54·9	507·3	220·0	545·9	3·47	351·8	−89·6	R

(d) Sal

No.	Type	N	D	I	K_0	K_{100}	K_{200}	K_{max}	θ_{63}	ϕ'	θ'	Polarity
1	I	3	201·4	−22·5	71·4	193·1	209·2	236·8	5·26	257·7	−68·6	R
2	I	3	214·0	−32·4	2·4	3·5	2·9	10·8	24·67	240·4	−57·5	R
3	E	3	1·3	+15·1	32·7	35·3	70·8	141·9	6·80	148·9	+80·9	N
4	E	3	183·2	+ 5·2	1·3	3·3	3·7	3·8	—	—	—	*
5	E	3	359·4	−16·8	5·7	24·6	107·1	126·2	7·13	158·5	+64·7	N
6	I	4	215·9	+69·2	23·8	1·8	1·7	16·5	19·96	315·7	−13·9	T
7	I	4	3·0	+ 6·0	1·5	2·7	3·1	3·2	—	—	—	*
8	I	4	313·3	+ 5.7	2·2	1·8	1·8	1·9	—	—	—	*

(e) Maio

No.	Type	N	D	I	K_0	K_{100}	K_{200}	K_{max}	θ_{63}	ϕ'	θ'	Polarity
1	E	4	301·7	+31·4	23·1	18·5	1·5	20·6	17·84	257·3	+34·2	N
2	I	5	6·1	− 4·6	208·4	203·9	147·2	382·0	4·14	137·4	+71·5	N

Table 1 (*continued*)

No.	Type	N	D	I	K_0	K_{100}	K_{200}	K_{max}	θ_{63}	ϕ'	θ'	Polarity
3	I	4	12·5	+ 2·8	22·7	125·2	227·7	452·4	3·81	113·9	+71·5	N
4	E	5	178·2	−16·4	40·9	129·9	215·9	225·4	5·39	351·5	−83·0	R

(f) Sao Tiago

No.	Type	N	D	I	K_0	K_{100}	K_{200}	K_{max}	θ_{63}	ϕ'	θ'	Polarity
1	E	3	346·0	+14·2	242·1	391·7	261·7	490·9	3·66	218·0	+74·2	N
2	E	3	349·8	+12·6	101·2	126·9	120·8	184·4	5·96	206·3	+76·7	N
3	E	3	4·9	+ 5·6	173·7	260·4	104·4	280·2	4·84	134·3	+76·8	N
4	E	3	0·0	+31·0	281·1	414·1	391·6	521·0	3·55	335·6	+88·4	N
5	E	3	357·7	+ 3·3	237·0	513·1	789·0	1506·6	2·09	166·0	+76·4	N
6	E	3	328·4	+ 0·4	1·9	4·5	4·8	4·8	—	—	—	*
7	E	3	25·7	+37·3	50·8	101·0	64·5	331·7	0·80	49·3	+64·9	N
8	E	3	13·6	+23·1	172·2	311·0	301·6	369·2	4·22	77·8	+76·5	N
9	E	3	26·7	+42·3	770·5	489·1	448·0	531·1	3·51	41·8	+63·3	N
10	E	3	3·1	+14·9	829·9	7202·5	489·1	7335·1	0·95	134·0	+81·9	N
11	E	3	5·5	+25·4	466·3	288·1	325·2	750·9	2·96	84·1	+84·4	N
12	E	3	15·7	+24·0	354·8	417·3	256·5	465·6	3·75	73·9	+74·5	N
13	E	2	356·2	+24·6	250·6	709·2	894·9	2305·7	1·69	216·0	+85·6	N
14	E	3	205·5	−03·8	4·8	1·9	2·1	2·4	—	—	—	*
15	E	3	161·8	−38·0	2·6	2·0	3·0	3·1	—	—	—	*
16	E	3	7·5	−10·9	1·1	2·3	2·6	2·8	—	—	—	*
17	E	3	352·0	+ 4·8	18766·0	180·3	104·5	2051·2	1·79	188·9	+75·1	N
18	E	3	336·5	+ 5·9	6·0	5·6	5·6	5·8	33·58	221·0	+63·9	N
19	E	3	355·8	+ 8·7	107·6	124·5	150·4	158·0	6·44	177·7	+78·5	N
20	E	3	354·0	+ 6·7	163·4	1177·9	1146·8	2688·1	1·56	183·8	+76·8	N
21	E	3	356·2	+ 7·6	11·4	52·6	161·7	161·3	6·38	175·0	+78·1	N
22	E	3	345·3	+30·6	109·8	120·1	143·2	166·4	6·25	254·0	+75·8	N
23	E	3	320·3	+ 5·1	2·1	1·7	1·5	2·0	—	—	—	*
24	E	3	343·4	+ 3·1	363·0	227·9	503·3	520·4	3·55	208·5	+68·7	N
25	E	3	344·2	+ 9·8	43·7	35·5	35·5	41·4	12·59	214·9	+71·4	N
26	E	3	355·4	+ 2·2	1040·5	511·4	277·6	944·5	2·64	174·7	+75·3	N
27	E	3	356·7	+11·6	475·5	702·0	2321·0	2716·0	1·55	176·1	+80·2	N
28	E	3	184·1	−12·6	211·3	180·0	232·8	290·8	4·75	311·0	−80·4	R

Table 1 (*continued*)

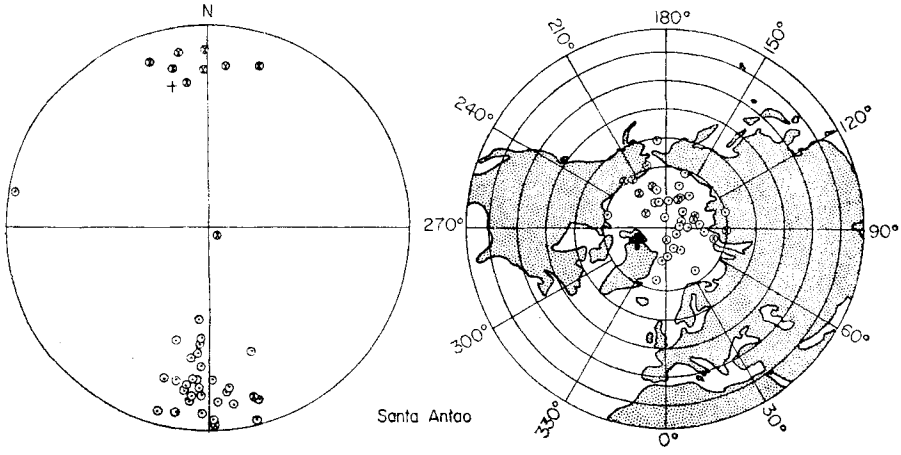
No.	Type	N	D	I	K_0	K_{100}	K_{200}	K_{max}	θ_{63}	ϕ'	θ'	Polarity
29	E	3	182·2	−10·9	100·8	132·2	233·8	235·6	5·28	323·4	−80·2	R
30	E	3	178·1	−45·1	1·2	1·5	1·8	2·2	—	—	—	*
31	E	3	188·3	−24·0	1134·0	614·8	535·1	779·8	2·90	263·0	−81·6	R
32	I	7	164·8	−28·2	22·9	73·4	75·8	124·9	7·25	68·1	−75·3	R
33	E	3	165·0	−15·6	103·1	355·2	277·9	370·4	4·21	42·0	−73·7	R
34	E	3	183·4	−26·6	206·0	340·6	198·5	533·7	3·51	263·4	−86·5	R
35	I	3	161·4	−13·7	50·5	4·2	248·6	246·3	5·16	44·3	−70·0	R
36	E	3	183·1	−24·4	149·1	198·6	177·8	250·8	5·12	283·9	−86·2	R

(g) Fogo

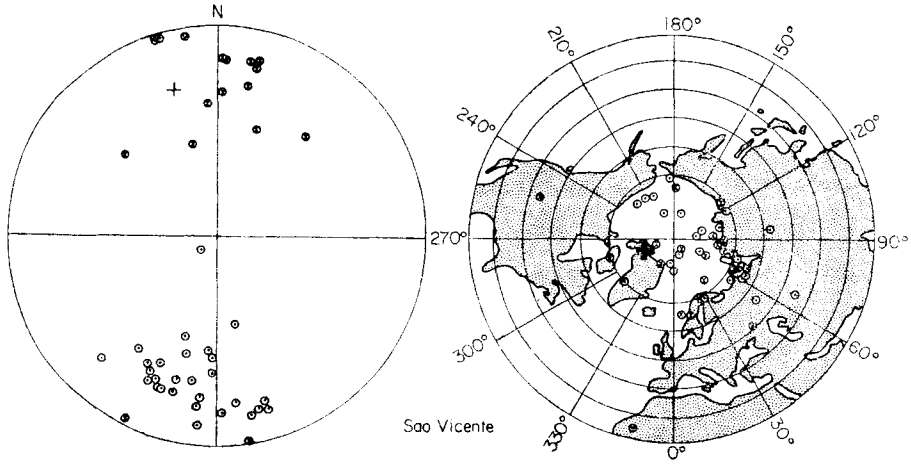
No.	Type	N	D	I	K_0	K_{100}	K_{200}	K_{max}	θ_{63}	ϕ'	θ'	Polarity
1	**E**	**8**	358·2	+ 0·8	225·3	275·1	165·1	296·1	4·7	162·7	+75·2	N

Notes on Table 1:

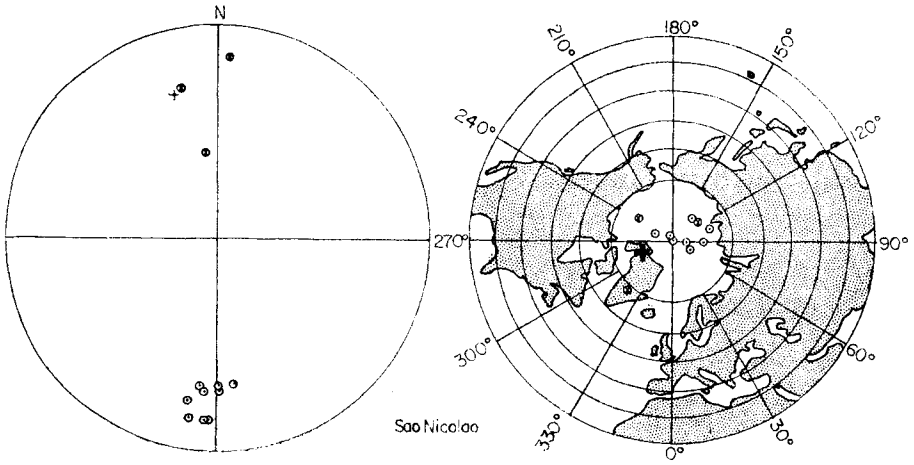
No.=number of sampled body, corresponding to sample location as indicated in Figs 2–8. Type=either extrusive (E) or intrusive (I). N=number of cores per body. $D°$ and $I°$ are declination and inclination of NRM east of north, and with respect to horizontal respectively (+ is above horizontal; − is below horizontal). K=Fisher (1953) precision parameter (suffixes, 100 and 200, refer to demagnetizing field value). K_{max} =Fisher precision parameter obtained by taking one demagnetized NRM value from each specimen so that K is maximum (scatter of NRM directions minimum). θ_{63}=the circular standard deviation. ϕ' and θ' are longitude and latitude of virtual geomagnetic pole (it is stressed that the virtual geomagnetic pole is not implied to be the exact magnetic or geographic pole position, which cannot be expected to be defined using only one body; it is merely plotting convention, being the surface expressions of the hypotihetical geocentric dipole which would give rise to the observed NRM). ϕ' is in degrees east of north; θ' is negative for southern latitudes. Polarity is either N (normal), R (reversed), or T (transitional, or equatorward of 30° latitude). The D, I, θ_{63}, ϕ' and θ' results corresponds to the K_{max} data, and result from unit vector per core.


N
180°
210°
150°
240°
120°
270°
90°
300°
60°
330°
30°
0°
Santa Antao

(a)


N
180°
210°
150°
240°
120°
270°
90°
300°
60°
330°
30°
0°
Sao Vicente

(b)


N
180°
210°
150°
240°
120°
270°
90°
300°
60°
330°
30°
0°
Sao Nicolao

(c)

FIG. 3

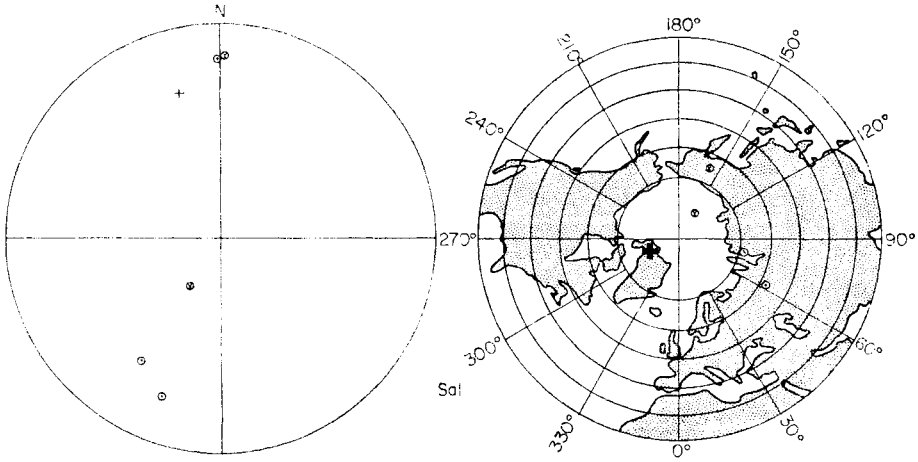


FIG. 3(d)

1000 m of volcanic material. The occurrence of interbasaltic soils suggests that the former possibility is more likely.

Sao Vicente. (Figs 2(b) and 3(b)). The highest point in Sao Vicente is less than one half that of the highest peak in Santa Antao, but more polarity variation is encountered. Limited sequences of lavas at the higher altitudes in Sao Vicente, such as units 9–13, 14–20, 36–42 and 47–51 are similar to the situation in Santa Antao, however, in that only reversed polarities occur. It is therefore possible that the period of reversed geomagnetic polarity recorded on Santa Antao was also a time of volcanic activity on Sao Vicente, although the lavas on the latter island are generally different in character, being much thinner, averaging only about 12 m in thickness, with very thick lavas absent. Sao Vicente also features petrologically unusual bodies such as the Pedras Blancas quartz monzonite plug (Part 1950b) which is intruded by narrow limburgite dikes. Both dike and plug (units 7 and 8) are reversely magnetized.

The lower slopes of the higher features on Sao Vicente are stratigraphically distinguishable from the upper areas, by their normal polarity (units 21–27, 43–45) or mixed polarities (units 28–36). Clearly, within-island stratigraphic mapping can

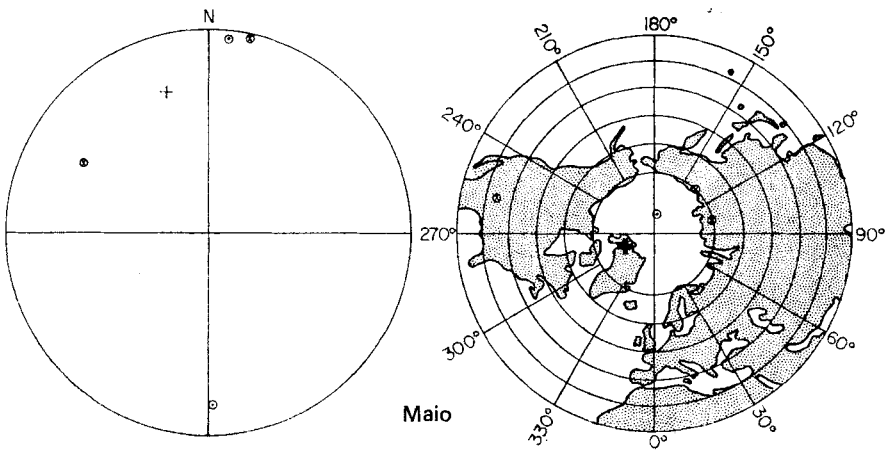


FIG. 3(e)

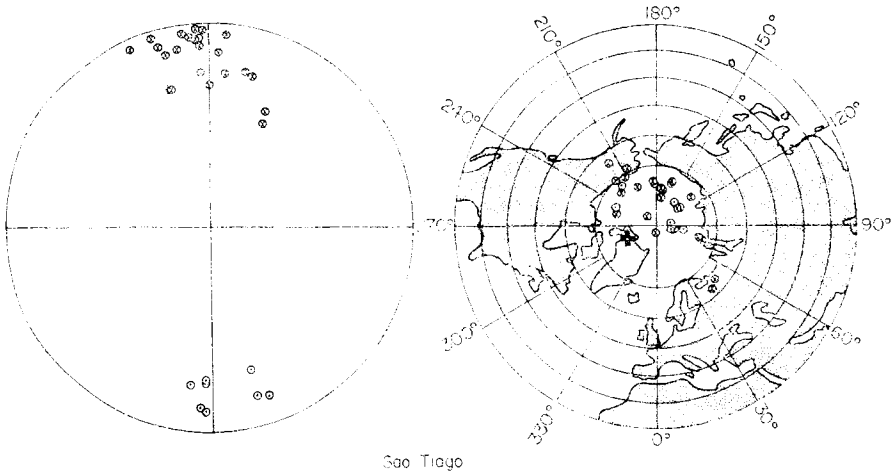


FIG. 3(f)

Fig. 3. Mean NRM directions for each body sampled on the islands of (a) Santa Antao, (b) Sao Vicente, (c) Sao Nicolao, (d) Sal, (e) Maio and (f) Sao Tiago. Presentation uses two conventions in each case: equal area stereogram (left) and polar projection (right). Explanation of symbols as follows: on steoreogram ⊙ =upper hemisphere, ⊗ =lower hemisphere, + =present geomagnetic field direction, N=geographic north; on polar projection ⊙ =south seeking pole, ⊗ =north seeking pole, + =present geomagnetic pole position. All data are transposed to the northern hemisphere.

profitably employ palaeomagnetic data in Sao Vicente. Watkins *et al.* (1966b, p. 473) have summarized the field methods used.

Sao Nicolao. (Figs 2(c) and 3(c)). A major erosional scarp immediately north of the town of Brava is composed of lavas in excess of 15 m thickness (units 1–5). These are of reversed polarity throughout. The limited sequence of lavas on the coast at Pedra do Guizinho (units 9–13) are indistinguishable from the Brava section on the basis of polarity, although normal polarity lavas exist between these areas (units 6–8). Very large outcrops exist in the western part of the island, but local circumstances prevented any visits to the area.

Sal. (Figs 2(d) and 3(d)). The reversed polarity of two topographically prominent columnar basalt piles or plugs (units 1 and 2) suggest that they may not be separate in time, but they differ in time of emplacement from the normal polarity phonolite masses (units 7 and 8) and the prominent lava near Palmeira (unit 5).

In general, drastic weathering characterizes any other positive relief in Sal, and those volcanics below the major unconformity cannot be readily related to their original *in situ* position. For these reasons, only limited palaeomagnetic data has been obtained for the island, and this is interpreted to be unlikely to represent an axial dipole, because of limited cancellation of the secular variation components.

Maio. (Figs 2(e) and 3(e)). Field operations were extremely difficult on Maio, for several reasons. In any case, the island resembles Sal in that weathering has virtually eliminated all but a few suitable outcrops above the major Post-Upper Cretaceous unconformity and, therefore, no axial dipole is likely to result from the data.

A monzonite intrusion (unit 2) with very large tourmalines is itself intruded by limburgite dikes (unit 3). Both units are of normal polarity. An extensive horizontal pillow lava (unit 4) underlying a thick organic limestone has a reversed polarity which contrasts with the normal polarity of the other sampled lava (unit 1), thus providing a potential in local mapping techniques.

Table 2

Cape Verde Islands palaeomagnetic data
(Group analysis)

Group	B	N	α_{95}	dp	dm	K	D	I	θ_{63}	ϕ'	θ'
(a) Santa Antao	43(3)	129(9)	6·5	3·7	6·9	12·9	358·7	+22·7	22·5	168·2	84·5
(b) San Vicente	52(6)	161(18)	6·4	3·7	6·9	12·0	6·7	+26·8	23·4	87·1	83·0
(c) San Nicolau	13(1)	39(3)	7·2	4·2	7·8	37·3	1·7	+26·2	13·3	124·7	86·8
(d) Sal	8(3)	27(11)	46·5	23·2	46·5	3·7	15·3	−00·7	42·3	114·1	+67·3
(e) Maio	4(0)	18(0)	41·5	21·6	42·3	5·9	352·6	+12·8	33·5	197·2	+78·7
(f) San Tiago	36(6)	111(18)	5·2	2·8	5·3	26·8	357·4	+17·1	15·6	178·9	+82·3
(g) Fogo	1(0)	8(0)	3·2	1·6	3·2	296·1	358·2	+00·8	4·7	161·7	+75·2
All data combined	157(19)	491(59)	3·5	2·0	3·7	12·6	1·6	+22·2	22·8	136·6	+85·2

Notes on Table 2:

B = number of bodies sampled. N = total number of cores in B. Numbers in brackets after B figures refer to the number of bodies which do not provide an NRM direction with sufficient precision (Vincenz & Bruckshaw 1966) for further statistical analysis. Number in brackets after N figures corresponds to number in brackets after B figures. α_{95} = semi-vertical angle of 95 per cent confidence cone. dp and dm are angular lengths of 95 per cent confidence oval, parallel and normal to great circle through sample site and virtual geomagnetic pole position (ϕ', θ'). Other symbols as in Table 1. All statistics are Fisher (1953) statistics, applied using unit vector per body. Note that Foga data omitted from combined analysis.

Sao Tiago. (Figs 2(f) and 3(f)). From close to sea level at Ribeira da Barca, through deeply dissected valleys to the higher central part of the island to south of Santa Catarina, only normal magnetic polarities were reliably defined (units 1–16), although units 14 and 15 *may* be reversely magnetized. These lavas, which are up to 30 m in thickness, are succeeded southward by a sequence of thinner lavas, or flow units (units 17–27) which are also normal polarity throughout. A change southward to less rugged relief occurs in the area of unit 28, as the surface volcanics become generally younger, and it is south of this area that the sampled bodies are seen to be all of reversed polarity (units 28–36). This characteristic polarity includes that of the nepheline monzonite of Riberia Sao Jorge (unit 32), which has been described by Part (1950b), who has also described the petrology of some of the Miocene sediments near Praia, (Part 1951).

It remains to be seen whether the present simple division of the island into dominantly normal polarity central part, and reversed polarity southern segment, is applicable in general.

Fogo. Only an extremely short landing was possible on the island, although the morphology was examined in some detail from the air. A single lava on the northeast coast at latitude 24° 20′ West was sampled. This body is undoubtedly of Recent to Pleistocene age, and since the geomagnetic field is thought to have been entirely of normal polarity since 0·7 million years ago (Doell *et al.*, 1966), the lava was expected to be of normal polarity, and this is the case (Table 1(g)).

Sea cliffs up to 70 m high occur on the western and northwestern side of the island, however, and it is conceivable that rocks older than 0·7 million years, and therefore of possible reversed polarity, occur on the island.

(ii) *Combined data for each island*

Fig. 4 shows the virtual geomagnetic poles (VGP) obtained using unit vector per lava, for each island. Since only one sample was obtained from Fogo, the virtual geomagnetic pole is not included.

The VGP for Maio and Sal are furthest from the geographic pole, and this could be interpreted to indicate a greater age for the outcrops. But as argued earlier, the data from Sal and Maio are far too limited to permit consideration of the corresponding mean VGP as being palaeomagnetic poles (which, by convention, are taken to indicate a reasonable estimate of the geographic pole). This limitation is emphasized by the very large confidence ovals for Sal and Maio (Table 2) which easily enclose the geographic pole. Clearly the VGP for Sal and Maio can have little significance.

The VGP for Santa Antao, Sao Vicente, Sao Nicolao and Sao Tiago are interpreted, on the basis of data volume, to be reasonable representations of axial dipole configurations. The ovals of confidence (Table 2), which are omitted from Fig. 4, almost (but not quite) enclose the present geographic pole. This observation is consistent with Doell & Cox's (1962, p. 294) observation that there is some indication in the literature that the Miocene geographic pole may be slightly removed from the present geographic pole.

The relatively low inclination of the mean NRM values for each island (Table 2) indicate that the data are sensitive to rotation about a vertical axis. (This would not be the case with steep inclinations, as found in higher latitudes.) The high latitudes of the VGP show that no significant rotation of any island about a vertical axis has taken place since the Miocene. Therefore if any crustal spreading has taken place since the Miocene in this part of the Atlantic, it has not been of the differential type, which has been suggested to have occurred, for example, in the northeastern Pacific (Vine 1966), and perhaps in the Bay of Biscay (Watkins & Richardson 1967).

(iii) *Combined data for the entire survey*

All data, excluding that single data point from Fogo (which is clearly Post-Miocene) have been combined using unit vector per body, to produce the mean virtual

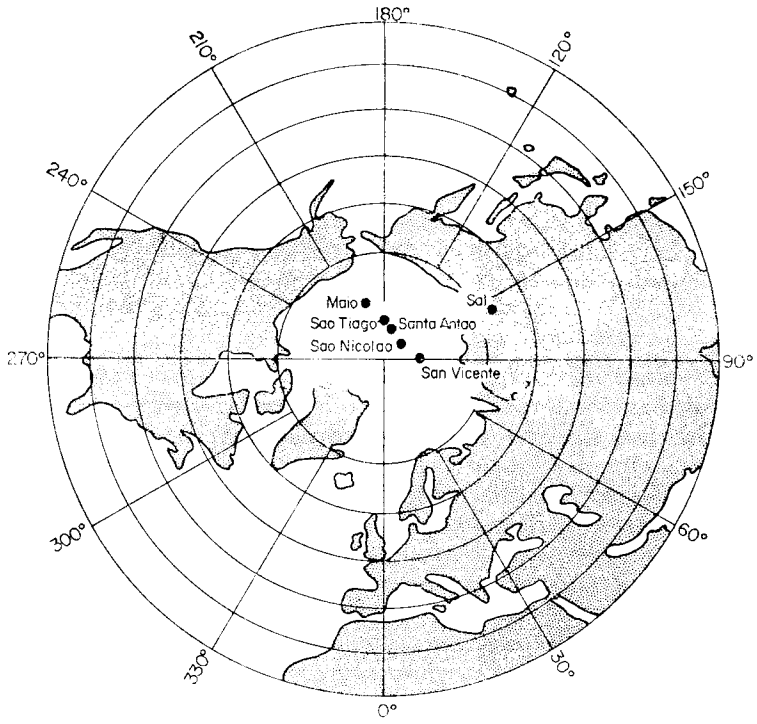


Fig. 4. Virtual geomagnetic poles for each island, obtained by assigning unit vector to each body. Ovals of confidence are omitted for clarity but are included in Table 2.

geomagnetic pole for the entire survey. The result is given in Table 2 and Fig. 5, which also includes results from earlier surveys of the Macaronesian insular region (Watkins *et al.* 1966a, b). The result is interpreted to be a good measure of the Miocene geomagnetic pole position for this part of the Atlantic. At the 95 per cent confidence level this is significantly different from the present spin axis, but is very similar to earlier reported Miocene pole positions for Madeira and some of the Canary Islands, suggesting similarity in average ages. It may be important to realize that identical field and laboratory methods have been used in all these studies. This reinforces the conclusions made when considering the results for each island: the Miocene is likely to be a period when the spin axis was slightly removed from its present position for a substantial period of time, and no Post-Miocene differential crustal spreading has occurred in the East–Central Atlantic. In addition, the combined results shown in Fig. 5 would appear to provide very strong evidence that, since the Lower Tertiary, the geographic pole has migrated towards its present position along a path confined to the area between 90° and 180° East, unless the geomagnetic field behaviour was controlled by unknown factors which vary regionally.

Summary and conclusions

Just less than five hundred separate cores from the Cape Verde Islands have been used to obtain the following results:

(1) Northeastern Santa Antao and Sao Vicente, which reach 1000 m above sea level, are composed dominantly of reversely magnetized lavas and, because of the existence of interbasaltic soils, this would appear to be most likely caused by a relatively long-duration reversed geomagnetic polarity epoch during the Miocene. Dagley *et al.* (1967, p. 28) believe that a long reversed polarity epoch existed during

the Lower Tertiary. Inspection of Irving's (1964) summary of palaeomagnetic data shows that many Miocene surveys report reversely magnetized materials, thus strengthening the suspicion of the existence of a long reversed polarity epoch during the Miocene. This would be in conflict with the polarity epoch duration predictions by Heirtzler *et al.* (1968, Fig. 3), which are based on constant rate of crustal spreading, and therefore would suggest that perhaps an underlying assumption involved in the predictions is not valid. On the other hand, thick sections of basalt have been observed to accumulate in a relatively short period: for example, Doell & Cox (1961) showed that much of the surface of the island of Hawaii accumulated during the last 0·7 million years, so that until additional relevant data is obtained, the possibility of rapid accumulation of the reversely magnetized sections on Santa Antao and Sao Vicente cannot be positively precluded.

(2) Polarity variations have high potential for future detailed stratigraphic mapping on most islands, but would appear to be of minimal value on most of northeastern Santa Antao and Sao Vicente (except at lower altitudes). Sao Tiago may possess a relatively simple palaeomagnetic stratigraphic character with the north-central area being of normal polarity, and the south of reversed polarity. On a between-island basis, Sao Tiago is strongly differentiated from the islands to the north by its normal polarity dominance.

(3) The palaeomagnetic poles for Santa Antao and Sao Vicente are similar to those from Sao Nicolao and Sao Tiago, the sampled lavas of which are mainly Miocene in age. Therefore the sampled lavas on Santa Antao and Sao Vicente are inferred to also be mainly Miocene.

(4) Palaeomagnetic poles for Santa Antao, Sao Vicente, Sao Nicolao and Sao Tiago, and the palaeomagnetic pole for the entire survey provide evidence that the

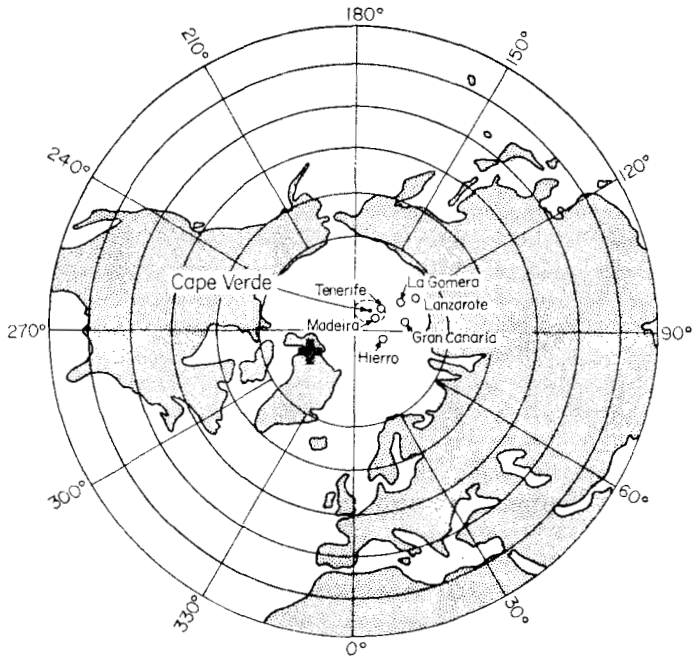


Fig. 5. Mean virtual geomagnetic pole for the entire survey, obtained by assigning unit vector per lava, with the 95 per cent confidence oval included. Heavy cross indicates the position of the present geomagnetic pole. Virtual geomagnetic poles from earlier surveys of the Macaronesian insular region (Watkins 1966a, b) are also shown, but 95 per cent confidence ovals are omitted for clarity.

Miocene palaeomagnetic pole, and therefore probably the spin axis, was slightly removed from the present geographic pole; that Post-Miocene tectonic rotation about a vertical axis (or differential crustal spreading) has not taken place in this part of the eastern Atlantic; and that the Miocene palaeomagnetic pole position for the Cape Verde Islands coincides with the established polar wandering curve for Europe.

(5) The sampled units on Maio and Sal are too few, and possess too poorly defined NRM directions, to provide conclusions other than the fact that both normal and reversed polarities occur, and therefore that magnetic data has local stratigraphic potential.

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Norman D. Watkins:
Department of Geology,
Florida State University,
Tallahassee,
Florida 32306.

A. Richardson and R. G. Mason:
Department of Geophysics,
Imperial College,
London, S.W.7.

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References

Abdel-Monem, A., Watkins, N. D. & Gast, P. W., 1967. Volcanic history of the Canary Islands, *Trans Am. geophys. Un.*, **48,** 226 pp.

Bebiano, J. B., 1932. A geologia do Arquipelago de Cabo Verde, *Commun. Serv. Geol. Portugal*, **18,** 276 pp.

Bebiano, J. B. & Pires Soares, J. M., 1951. Notes on some supposed Senonian fossils from Sao Nicolau island (Cape Verde islands), *18th International Geological Congress* Report 14, pp. 186–189.

Berthios, L., 1950. Contribution à la connaisance lithologique de l'archipel du Cap-Vert. *Portugese Min. Colonias, Est.*, **7,** 194.

Dagley, P., Wilson, R. L., Walker, G. P. L., Watkins, N. D., Sigurgeirsson, T., Haggerty, S. E., Smith, P. J., Edwards, J., Ade-Hall, J. M. & Grasty, R. L., 1967. Geomagnetic polarity zones for Icelandic lavas, *Nature, Lond.*, **216,** 25.

Doell, R. R., & Cox, A., 1961. Palaeomagnetism of Hawaiian lava flows, *Nature, Lond.*, **192,** 645.

Doell, R. R. & Cox, A., 1962. Paleomagnetism, *Adv. Geophys.*, **8,** 221.

Doell, R. R. & Cox, A., 1965. Measurement of the remanent magnetism of igneous rocks, *U.S. geol. Surv. Bull.* 1203A, 32 pp.

Doell, R. R., Dalrymple, G. B., & Cox, A., 1966. Geomagnetic polarity epochs: Sierra Nevada Data 3, *J. geophys Res.*, **71,** 531.

Fisher, R., 1953, Dispersion on a sphere, *Proc. R. Soc. Lond.*, **A217,** 295.

Friedlaender, I., 1913. Beitrage zur Kenntniss der Kape-verdischen Inseln—Die Ergebnisse einer Studienreise im Sommer, 1912, *D. Reimer, Berlin,* 109 pp.

Hausen, H., 1956. Contributions to the geology of Tenerife, *Soc. Sci. (Finland) Comm. phys. Math.*, **18,** 271.

Heirtzler, J. R. & Hayes, D. E., 1967. Magnetic boundaries in the North Atlantic ocean, *Science*, **157,** 185.

Heirtzler, J. R., Dickson, G. O., Herron, E. M., Pitman, W. C., & Le Pichon, X., 1968. Marine magnetic anomalies, geomagnetic field reversals, and motions of the ocean floor and continents, *J. geophys. Res.*, **73,** 2119.

Hurley, P. M., de Almeida, F. F. M., Melcher, G. C., Cordani, V. G., Rand, J. R., Kawashita, K., Vandoros, P., Pinson, W. H. & Fairbairn, H. W., 1967. Test of continental drift by comparison of radiometric ages, *Science*, **157**, 495.

Irving, E., 1964. *Paleomagnetism and its Application to Geological Problems*, Wiley, New York, 399 pp.

McBirney, A. R. & Gass, I. G., 1967. Relation of oceanic volcanic rocks to mid-oceanic rises and heat flow, *Earth planet. Sci. Lett.*, **2**, 265.

Menard, H. W., 1965. Sea-floor relief and mantle convection, *Phys. Chem. Earth*, **6**, 630.

Part, G. M., 1950(a). Volcanic rocks from the Cape Verde Islands, *Bull. Br. Mus., Nat. Hist.*, **1**, (2), 27.

Part, G. M., 1950(b). Occurrence of nephaline-monzonite and allied types in the Cape Verde archipelago, *Geol. Mag.*, **87**, 421.

Part, G. M., 1951. Tertiary and Pleistocene sediments from the Cape Verde archipelago, *Geol. Mag.*, **88**, 65.

Pires Soares, J. M., 1948. Observations géologiques sur les îles du Cap-Vert, *Bull. Soc. geol. France*, **5**, 18, 383.

Pires Soares, J. M., 1954. Les sédiments meubles des îles du Cap-Vert; bref renseignements sur leur nature, granulométrie morphoscopie et teneur en magnétite, *19th International Geological Congress* Vol. 2, p. 334.

Pires Soares, J. M., 1956(a) Description d'une faunule de pertinides des îles de Santiago et de Sao Nicolau (archipel du Cap Vert), *Proc. Conf. int. Afric. Ocident.*, 6 (Sao Tome), **2**, 271.

Pires Soares, J. M., 1956(b). Sobre alguns exemplares de turritelas fosseis da ilha do Sao Nicolau (archipelago do Cabo Verde), *Proc. Conf. int. Afric. Ocident.*, **1**, 306.

Pitman, W. C. & Heirtzler, J. R., 1966. Magnetic anomalies over the Pacific–Antarctic ridge, *Science*, **150**, 485.

Richardson, A. & Watkins, N. D., 1967. Palaeomagnetism of Atlantic islands: Fernando Noroñha, *Nature, Lond.*, **215**, 1470.

Sousa Torres, A. & Pires Soares, J. M., 1946. Formacoes sedimentares do archipelago de Cape Verde—I Actualizacao de conhecimentos, *Portugal Junta Mis Geog. Inv., Col. Mem. Ser. Geol.*, **3**, 397.

Sousa Torres, A. & Pires Soares, J. M., 1950. A idade dos sedimentos do archipelago de Cabo Verde, *Conf. Int. Afric. Ocident.*, **2** (Bissau, 1947) **1**, 85.

Stahlecker, R., 1935. Neocom auf der Kapverden-Insel Maio, *Neus Jahrb. Min. Geol. Palaeont.*, **B73**, 265.

Torre de Assuncao, C. F., 1954. Expedicao cientifica a ilha do Fogo; estudos petrograficos (1952–1953): *Portugal Junta Inv. Ultramar., Ser. Petrog.*, **1**, 156 pp.

Vincenz, S. A. & Bruckshaw, J. McG., 1960. Note on the probability distribution of a small number of vectors, *Proc. Camb. phil. Soc. Math. Phys. Sci.*, **56**, 21.

Vine, F. J., 1966. Spreading of the ocean floor: new evidence, *Science*, **154**, 1405.

Vogt, P. R. & Ostenso, N. A., 1967. Steady state crustal spreading, *Nature, Lond.*, **215**, 810.

Watkins, N. D., 1965. Paleomagnetism of the Columbia Plateaus, *J. geophys. Res.*, **70**, 1379.

Watkins, N. D., Richardson, A. & Mason, R. G., 1966(a). Paleomagnetism of the Macaronesian insular region: The Canary Islands, *Earth planet Sci. Lett.*, 225.

Watkins, N. D., Richardson, A. & Mason, R. G., 1966(b). Palaeomagnetism of the Macaronesian insular region: Madeira, *Earth planet. Sci. Lett.*, **1**, 471.

Watkins, N. D. & Richardson, A., 1967. Palaeomagnetism of the Lisbon volcanics, *Geophys. J. R. astr. Soc.*, **15**, 287.

Wilson, J. T., 1963(a). Pattern of uplifted islands in the main ocean basins, *Science*, **139**, 592.

Wilson, J. T., 1963(b). Hypothesis of the Earth's behaviour, *Nature, Lond.*, **198**, 925.